Eco-evo dynamics

Learning objectives

1. Understand the differences between GxE, phenotypic plasticity, local adaptation, and evolutionary rescue
2. Know how we detect each of these processes

last lecture: previously, we've assumed all individuals of a species are basically equal, all react to environment or species interactions in same way

eco-evo dynamics: addresses population dynamics when there's genetic variation within populations; composed of individuals of different genotypes, not all genetically the same

Genotype x environment interactions

Phenotypic plasticity vs. GxE

- Phenotypic plasticity - traits shift with environment within a genotype
- GxE - trait or fitness shifts with environment depend on genotype *different genotypes have different responses*

Plot soil moisture x biomass with different lines for different genotypes, ask if plasitcity or GxE with each one

plot1: No slope - no phenotypic plasticity, no GxE (what we assumed prior to this)

- reaction norm - plant response to a range of environmental conditions

plot2: Parallel lines with different intercepts - plasticity with G and E but not GxE

non-zero slopes mean biomass is shifting with enviornment (plasticity)

different intercepts mean genotypes differ in biomass, but shift with environment is same, no GxE

plot3: Slopes differ but don't cross - plasticity with GxE

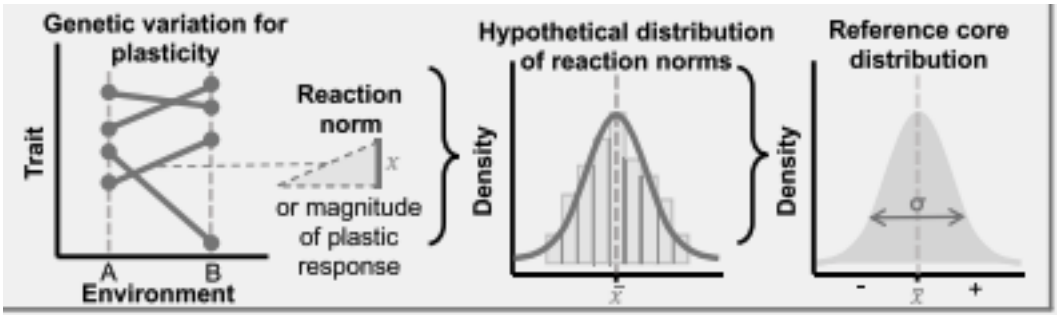
plot4: Slopes differ and cross - plasticity with GxE

Methods

- Common gardens - genotypes planted in same environment

draw multiple example plots with 4 genotypes planted in grid

- Reaction norms across space - measure trait response, create distribution, use as reference distribution vs. range edges



Local adaptation

main difference between this and the others is this has to do with fitness, not trait shifts, and comparison of fitness in source environment vs. other environments

- "adaptive" = affects fitness *but terminology is contentious, local adaptation has narrow definition*
  - can't confirm local adaptation from 1 life stage
- local adaptation - fitness highest in source environment; adapted to local conditions

Plot environment A & B by fitness

Plot1: Local adaptation - a line high in cold, low in warm; b line high in warm low in cold

plot2: GxE and local adaptation - a line higher than B line, decreasing between cold and warm, a more adapted

question: what would maladptation or the opposite of local adaptation look like?

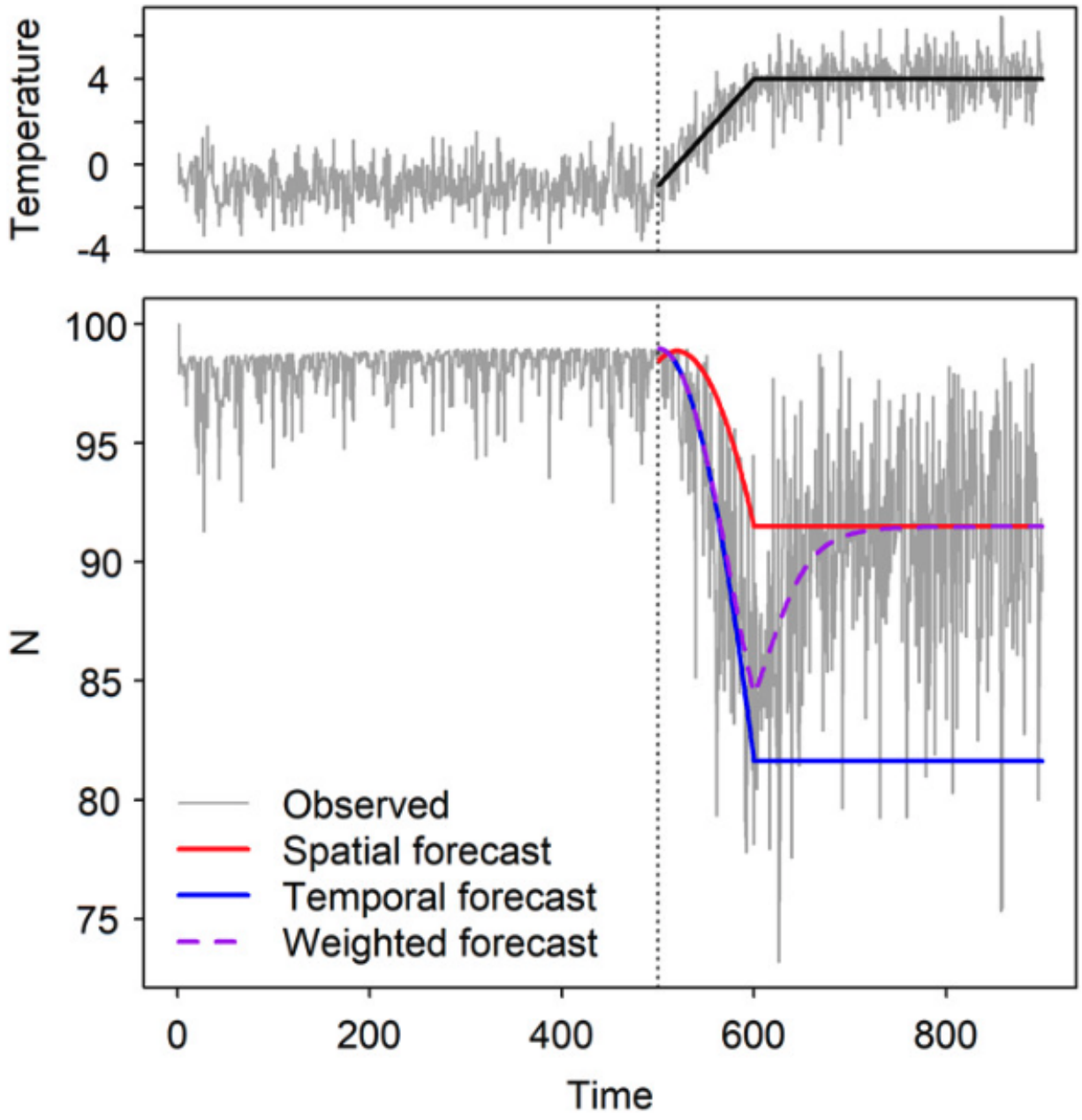
plot3: Maladaptation - opposite of local adaptation

Methods

- Reciprocal transplant - measure fitness in source vs. other environment for each species *this is what I drew above*
- Common garden - can be range of environments *change plot 1 x axis to continuous temperature axis*
  - multiple common gardens, each containing genotypes from other environments

Evolutionary rescue

- evolutionary rescue: when populations persist under climate stress *(or other stresses)* due to rapid evolution
  - rapid evolution (rapid adaptation): genotype frequencies within a population change
  - mechanisms: genetic variation & dispersal *less commonly: mutations*
  - examples: crops evolving resistance to pesticides, antibiotic resistant bacteria


Temperature
4
0
-4
N
100
95
90
85
80
75
Observed
Spatial forecast
Temporal forecast
Weighted forecast
0
200
400
600
800
Time